

# Factor-Graph Aeromagnetic Compensation for Airborne Magnetic-Anomaly Navigation

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**Abstract**—Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft’s own field must be compensated online when no calibration flight is available (the cold-start regime). A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We show that a factor-graph formulation reaches the same accuracy without a neural network by carrying the TL coefficients as graph variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight. Across five long survey lines from three flights and two maps, the smoother is the best of three cold-start estimators on every full-length line (8 of 8 cases): it stays bounded where a plain online-TL EKF diverges, and it beats a reimplemented EKF+TL+NN—which the neural network keeps bounded—without using a neural network. We also give an exact linear condition for the joint observability of compensation and navigation, from which an exogeneity design rule for the compensation features follows.

**Index Terms**—Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.

## NOMENCLATURE

|                        |                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{x}_t$         | INS error state at epoch $t$ (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$ . |
| $\beta_t$              | Online Tolles–Lawson compensation coefficients.                                                                                         |
| $\chi_t$               | Joint state $(\mathbf{x}_t, \beta_t)$ .                                                                                                 |
| $\Phi_t, \mathbf{Q}_t$ | Pinson state-transition matrix and process covariance.                                                                                  |
| $z_t, \eta_t$          | Scalar magnetometer measurement and its noise, variance $R$ .                                                                           |
| $h(\cdot)$             | Anomaly-map interpolation (with IGRF core) as a function of position.                                                                   |
| $\mathbf{g}_t$         | Map gradient $\partial h / \partial \mathbf{p}$ at epoch $t$ .                                                                          |
| $\mathbf{A}_t$         | Tolles–Lawson basis row from the three-axis fluxgate.                                                                                   |
| $c_t$                  | Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \beta_t$ .                                                                     |
| $L_w, L_o$             | Sliding-window length and overlap (look-ahead).                                                                                         |
| $\rho, \delta$         | Robust kernel and its Huber threshold.                                                                                                  |
| $\mathbf{G}, \Psi$     | Stacked map-gradient and TL-regressor matrices over a window.                                                                           |

## I. INTRODUCTION

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Reproducible code and experiments are provided in the MagNav.jl research branch `claude/fgo-problem-research-bllgbr`.

**I**NERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, difficult to deny or spoof, and observable with a lightweight scalar magnetometer [3], [4]. The same idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [5], [6], [7]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft’s own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar magnetic field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The public benchmark that has catalyzed recent work is the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset, comprising flight-grade INS, multiple onboard magnetometers, and high-resolution anomaly maps [8]. That dataset also exposes the central obstacle: the dominant error source is the aircraft itself, not the map. The platform’s own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [9], which regresses the interference onto sixteen attitude-dependent terms—permanent, induced, and eddy-current—formed from a three-axis fluxgate magnetometer. The coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, where the anomaly signal is negligible and the interference dominates. Many operational scenarios, however, admit no such calibration flight: the coefficients drift with temperature and configuration, and a survey may begin with

no prior calibration at all. In this cold-start regime the compensation coefficients must be learned online and jointly with navigation, from the same measurements used to fix position—coupling that makes both estimation and its analysis substantially harder than the calibrated case.

A second difficulty, prominent on the SGL 2020 cabin magnetometers [8], is that the linear TL basis does not span the true interference: platform currents and other non-rigid, non-attitude sources contribute a large fraction of the field that no set of Tolles–Lawson coefficients can represent [10]. This has motivated a line of learning-augmented compensation on this dataset—neural-network heading-error and denoising models [11], [10], and the online EKF+TL+NN whose residual network mops up exactly this un-modeled remainder [12]. The scalar optically-pumped magnetometers used for the cabin channels add their own orientation-dependent errors—heading error and equatorial dead zones [13], [14]—on top of the platform field. How to absorb the non-TL remainder *without* a neural network is one of the questions this paper addresses.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [15] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [5] and nonlinear Kalman filtering for terrain-aided navigation [6] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [16], [17], [18], [7]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [19]. Beyond the EKF, unscented filtering [20], invariant formulations that respect the state manifold [21], and mixture/particle representations [22] improve consistency under nonlinearity. *However*, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [9], [23], [24]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [11], model-stability and robustness of the regression have been studied in gradiometry [25], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [10]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses [12]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and is, to our knowledge, the strongest reported cold-start result on that benchmark. *Nevertheless*, the network requires

a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and—being embedded in an EKF—inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [26] whose maximum-a-posteriori (MAP) solution is found by sparse non-linear least squares [27], [28], the same structure underlying bundle adjustment and graph SLAM [29], [30], [31]. Incremental and square-root smoothers—square-root SAM [32], iSAM [33], iSAM2 [34], and exactly-sparse delayed-state filters [35]—make the solution efficient enough for online use, and fixed-lag sliding-window optimization [36], [37] bounds its cost. On-manifold inertial preintegration [38], factor-graph inertial fusion [39], and modern visual-inertial estimators [40] have brought the approach firmly into navigation. *Despite these advances*, factor-graph estimation in navigation has been developed almost entirely for visual-inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks; it has not been brought to bear on *joint* online aeromagnetic compensation and map matching, and the observability of estimating a compensation model jointly with position from map matching has not been characterized.

In this work we cast online Tolles–Lawson compensation as a set of time-varying variables in a factor graph and solve the joint compensation-and-navigation problem as a fixed-lag sliding-window smoother. The per-window relinearization supplies the time-varying compensation that the competing method obtains from an online network, and deferring commitment over the window removes the causal filter’s failure mode. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [41], so smoothing per se is not the contribution; the joint windowed estimation of a time-varying compensation is. The paper makes the following contributions:

- 1) **NN-free online compensation:** we formulate online TL compensation as factor-graph variables in a fixed-lag sliding-window smoother, an alternative to online EKF+TL+NN that needs no neural network (Section III).
- 2) **Observability condition:** we prove an exact linear condition for the joint observability of compensation and navigation and derive an exogeneity design rule for the compensation features (Section IV).
- 3) **Reimplemented baseline and breadth study:** we reimplement—rather than merely cite—the online EKF+TL+NN result and evaluate both estimators on five long survey lines from three flights and two maps (Section VI).
- 4) **Sensor-error factors:** we cast physically motivated scalar-magnetometer error terms as graph variables that a position-only estimator cannot represent (Section III).

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the sliding-

window smoother and the compensation and sensor-error factors. Section IV establishes the observability condition. Section V describes the reimplemented EKF+TL+NN baseline, and Section VI reports the experiments. Section VII discusses the results and their limitations, and Section VIII concludes.

## II. PROBLEM FORMULATION

We first fix notation and state the estimation problem. Throughout, boldface lowercase denotes vectors and boldface uppercase matrices;  $\mathcal{N}(\cdot, \cdot)$  is the Gaussian density; and  $\|\mathbf{v}\|_{\mathbf{M}}^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$  is the squared Mahalanobis norm.

### A. INS error state and dynamics

We adopt the standard Pinson error model of a strapdown INS [2], [1], [42]. The error state

$$\mathbf{x} = [\delta \mathbf{p}^\top \delta \mathbf{v}^\top \boldsymbol{\psi}^\top h_a \hat{\mathbf{a}} \mathbf{b}_a^\top \mathbf{b}_g^\top S]^\top \in \mathbb{R}^{18} \quad (1)$$

stacks position error  $\delta \mathbf{p}$ , velocity error  $\delta \mathbf{v}$ , attitude error  $\boldsymbol{\psi}$  (three components each), the baro-inertial altitude-aiding pair  $(h_a, \hat{\mathbf{a}})$  that damps the vertically unstable channel, accelerometer and gyroscope biases  $\mathbf{b}_a, \mathbf{b}_g$ , and a scalar first-order Gauss–Markov (FOGM) state  $S$  that absorbs residual magnetic disturbance and map error. The discretized, linearized dynamics are

$$\mathbf{x}_{t+1} = \boldsymbol{\Phi}_t \mathbf{x}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \quad (2)$$

where  $\boldsymbol{\Phi}_t$  is the Pinson state-transition matrix (Appendix A) and  $\mathbf{Q}_t$  the process covariance; the FOGM block of  $\boldsymbol{\Phi}_t$  is  $e^{-\Delta t/\tau}$  with correlation time  $\tau$ .

### B. Scalar magnetometer measurement

The scalar magnetometer reads

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, R), \quad (3)$$

where  $h(\cdot)$  interpolates the crustal anomaly map (with the IGRF core field restored) at the aircraft position  $\mathbf{p}_t = \bar{\mathbf{p}}_t + \delta \mathbf{p}_t$  and  $c_t$  is the aircraft interference. Linearizing about the nominal position,

$$z_t \approx h(\bar{\mathbf{p}}_t) + \mathbf{g}_t^\top \delta \mathbf{p}_t + c_t + \eta_t, \quad \mathbf{g}_t = \left. \frac{\partial h}{\partial \mathbf{p}} \right|_{\bar{\mathbf{p}}_t}, \quad (4)$$

so the map gradient  $\mathbf{g}_t$  is the *sole* source of horizontal position information: where  $\mathbf{g}_t \approx \mathbf{0}$  (a locally flat map) position is momentarily unobservable, and everywhere the interference  $c_t$  competes with the  $\mathbf{g}_t^\top \delta \mathbf{p}_t$  term for the same measurement. The map  $h$  is stored as a gridded anomaly product upward-continued to the survey altitude; we evaluate it and its gradient by bicubic interpolation, which yields the continuous  $\mathbf{g}_t$  required to linearize (4), and we restore the IGRF core field so that  $z_t$  and  $h$  are on a common absolute scale. The gradient magnitude  $\|\mathbf{g}_t\|$  varies by more than an order of magnitude along a line, which is why position accuracy is not uniform in time even for a perfect compensator—a point the error traces of Section VI make visible.

### C. Tolles–Lawson interference model

Classical aeromagnetic compensation writes the interference as a linear combination of attitude-dependent basis terms [9], [23], [24],

$$c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t, \quad (5)$$

where  $\mathbf{A}_t \in \mathbb{R}^m$  is the TL basis row of (5) built from the three-axis fluxgate; its classical rank deficiency is a gauge freedom (Remark 2). With direction cosines  $\hat{\mathbf{b}}_t = [u \ v \ w]^\top$  of the Earth field in body axes and total field magnitude  $B_t$ , the permanent (3), induced (6), and eddy-current (9) groups are

$$\mathbf{A}_t = [\underbrace{u, v, w}_{\text{perm.}}, \underbrace{B_t u^2, \dots, B_t w^2}_{\text{induced}}, \underbrace{B_t u \dot{u}, \dots, B_t w \dot{w}}_{\text{eddy}}]^\top, \quad (6)$$

optionally augmented by a bias term, giving  $m \leq 19$  coefficients  $\boldsymbol{\beta}_t$ . In the calibrated setting  $\boldsymbol{\beta}$  is fit once and frozen; in the cold-start setting it is unknown at takeoff and drifts slowly, so we treat it as a time-varying state with a random-walk prior  $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{w}_t^\beta$ ,  $\mathbf{w}_t^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta)$ .

### D. Batch maximum-a-posteriori estimate

Collecting the navigation and compensation variables into  $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$  over a window of  $N$  epochs, the MAP estimate minimizes

$$J(\boldsymbol{\chi}) = \|\boldsymbol{\chi}_1 - \boldsymbol{\chi}_0\|_{\mathbf{P}_0^{-1}}^2 + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \boldsymbol{\Phi}_t \mathbf{x}_t\|_{\mathbf{Q}_t^{-1}}^2 + \sum_{t=1}^{N-1} \|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|_{(\mathbf{Q}^\beta)^{-1}}^2 + \sum_{t=1}^N \rho\left(\frac{r_t(\boldsymbol{\chi}_t)}{\sqrt{R}}\right), \quad (7)$$

with measurement residual  $r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t$  and  $\rho$  a possibly robust kernel; the prior term anchors both blocks of  $\boldsymbol{\chi}_1 = (\mathbf{x}_1, \boldsymbol{\beta}_1)$ , at cold start with  $\boldsymbol{\beta}$ -block  $\mathbf{P}_0^\beta$  (Table I). We make the standard modeling assumptions of aided inertial navigation [1], [22].

*Assumption 1:* The process and measurement noises  $\mathbf{w}_t, \mathbf{w}_t^\beta, \eta_t$  are zero-mean, white, mutually independent, and Gaussian; the linearizations (2) and (4) are valid over a window; and the prior  $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$  is Gaussian.

Assumption 1 is the same one under which the EKF and RTS smoother are derived, so any performance difference between the estimators reflects their *structure*, not their noise models. Under it the batch estimate coincides with the classical smoother.

*Lemma 1:* Under Assumption 1, with  $\rho(u) = \frac{1}{2}u^2$  and  $\boldsymbol{\beta}$  held fixed, the minimizer of (7) equals the Rauch–Tung–Striebel (RTS) smoother estimate over the window.

*Proof:* With  $\rho$  quadratic and  $\boldsymbol{\beta}$  fixed, (7) is a positive-definite quadratic in the Gaussian chain  $\{\mathbf{x}_t\}$ , so its minimizer is the posterior mean  $\mathbb{E}[\mathbf{x}_{1:N} \mid z_{1:N}]$ . For a linear–Gaussian state-space model that conditional mean is exactly what the forward Kalman recursion followed by the backward RTS recursion computes [41]; the two therefore coincide. ■

*Remark 1:* Lemma 1 is why we do not claim smoothing itself as a contribution: for a static, known compensation the batch graph and a filter–smoother pass are equivalent. The contribution is what the graph makes easy that a filter does

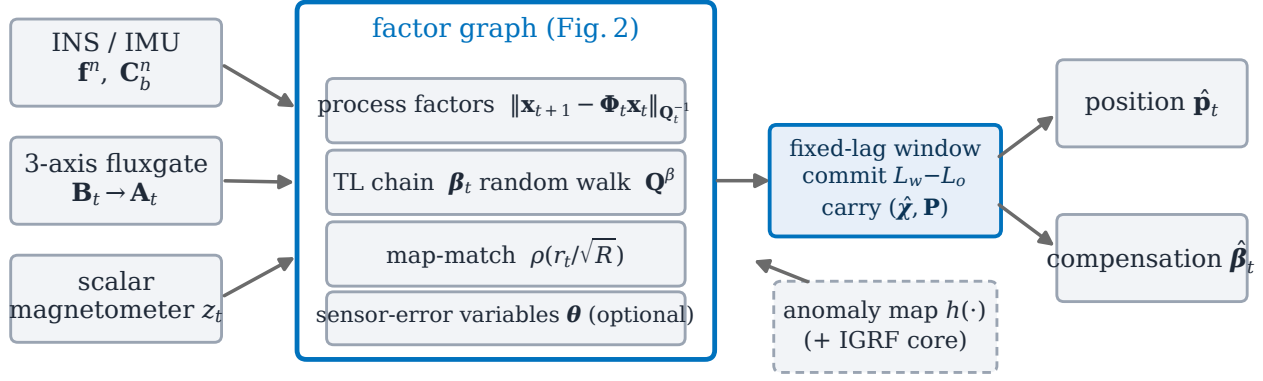


Fig. 1. Proposed estimator. The INS error state, the three-axis fluxgate (Tolles–Lawson basis), and the scalar magnetometer feed a factor graph whose variables are the navigation error, the online TL coefficients, and optional sensor-error states. A fixed-lag window solves the graph by robust Gauss–Newton/QR and outputs position and compensation; the anomaly map enters through the map-match factor.

not—jointly estimating the *time-varying*  $\beta$ , adding the non-Gaussian robust kernel of (10), and appending the sensor-error variables of Section III—together with the fixed-lag windowing that keeps it causal-latency bounded.

### III. METHOD

The proposed estimator (Fig. 1) represents navigation, on-line compensation, and optional sensor-error states as variables of a single factor graph, and solves it over a fixed-lag sliding window by robust Gauss–Newton. This section develops the factors, the windowing, the robust solver, and the sensor-error terms.

#### A. Online TL coefficients as graph variables

We append the compensation coefficients  $\beta$  to the state, so the measurement factor of (3) becomes, in residual form,

$$r_t(\chi_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \beta_t - S_t, \quad (8)$$

with Gaussian information  $R^{-1}$ , and its Jacobian row is

$$\mathbf{H}_t = \begin{bmatrix} \underbrace{\mathbf{g}_t^\top}_{\delta \mathbf{p}} & \underbrace{\mathbf{0}}_{\delta \mathbf{v}, \psi, h_a, \hat{a}, \mathbf{b}} & \underbrace{\mathbf{A}_t^\top}_{\beta} & \underbrace{1}_S \end{bmatrix}. \quad (9)$$

The window posterior over the joint states  $\chi_t = (\mathbf{x}_t, \beta_t)$  is the product of four factor families, drawn explicitly in Fig. 2:

- 1) a *joint prior* on  $\chi_1 - \mathbf{P}_0$  at cold start, or the smoothed mean and covariance carried from the previous window (Section III-B);
- 2) *Pinson–FOGM process factors*  $\|\mathbf{x}_{t+1} - \Phi_t \mathbf{x}_t\|_{\mathbf{Q}_t^{-1}}^2$ , parameterized by the INS mechanization (specific force  $\mathbf{f}^n$ , attitude  $\mathbf{C}_b^n$ ) through (12);
- 3) *compensation random-walk factors*  $\|\beta_{t+1} - \beta_t\|_{(\mathbf{Q}^\beta)^{-1}}^2$ , which set how quickly the compensation may adapt— $\mathbf{Q}^\beta \rightarrow \mathbf{0}$  recovers a single static  $\beta$ ;
- 4) *robust scalar-magnetometer factors*  $\rho(r_t/\sqrt{R})$  with the residual (8), each coupling  $\delta \mathbf{p}_t$  (through the map gradient  $\mathbf{g}_t$ ), the disturbance state  $S_t$ , and  $\beta_t$  (through the fluxgate regressor  $\mathbf{A}_t$ ).

The INS mechanization, the fluxgate (through  $\mathbf{A}_t$ ), and the anomaly map (through  $h$  and  $\mathbf{g}_t$ ) enter as *known factor parameters*, not as estimated variables; the optional sensor-error variables  $\theta$  of Section III-F attach to every measurement factor when enabled. Two implementation details: in the online estimator only the horizontal components of  $\mathbf{g}_t$  are used (the vertical channel is baro-damped and the map is evaluated at survey altitude), whereas the batch validation of Section VI-A uses the full three-component gradient; and the implementation orders the state vector as  $[\text{Pinson-17}; \beta; S]$ , a permutation of (1) that does not affect the estimate. Because consecutive coefficient blocks are chained rather than instantaneous, the compensation is identified from the whole trajectory, and because the batch solution uses future as well as past measurements, early navigation states benefit from calibration information that only becomes observable after later maneuvers—the property a causal filter lacks.

#### B. Fixed-lag sliding window

A single batch over a long line forces one static  $\beta$ , which underfits time-varying interference; a per-epoch filter, conversely, cannot use future data. We take the middle path and process the flight in overlapping windows of length  $L_w$  with overlap  $L_o$ , each a batch factor graph, in the manner of fixed-lag and incremental smoothing [34], [36], [37]. Window  $k$  spans epochs  $[s_k, s_k + L_w]$ ; it is solved as a batch, commits only its leading stride of  $L_w - L_o$  epochs, and carries the full smoothed state and covariance of the last committed epoch forward as a Gaussian prior for window  $k+1$  (Fig. 3). The trailing overlap  $L_o$  acts as bounded smoother look-ahead, so each committed estimate has seen  $L_o$  epochs of future data. Carrying the full covariance—not just the mean—keeps navigation continuous across window boundaries. One approximation deserves note: the carried prior is the *smoothed* state at the commit boundary, which has already seen the overlap measurements, and window  $k+1$  processes those  $L_o$  epochs again, so the overlap information is counted twice. An exactly consistent scheme would carry the forward-filtered state or marginalize the overlap; we accept the bounded optimism (the

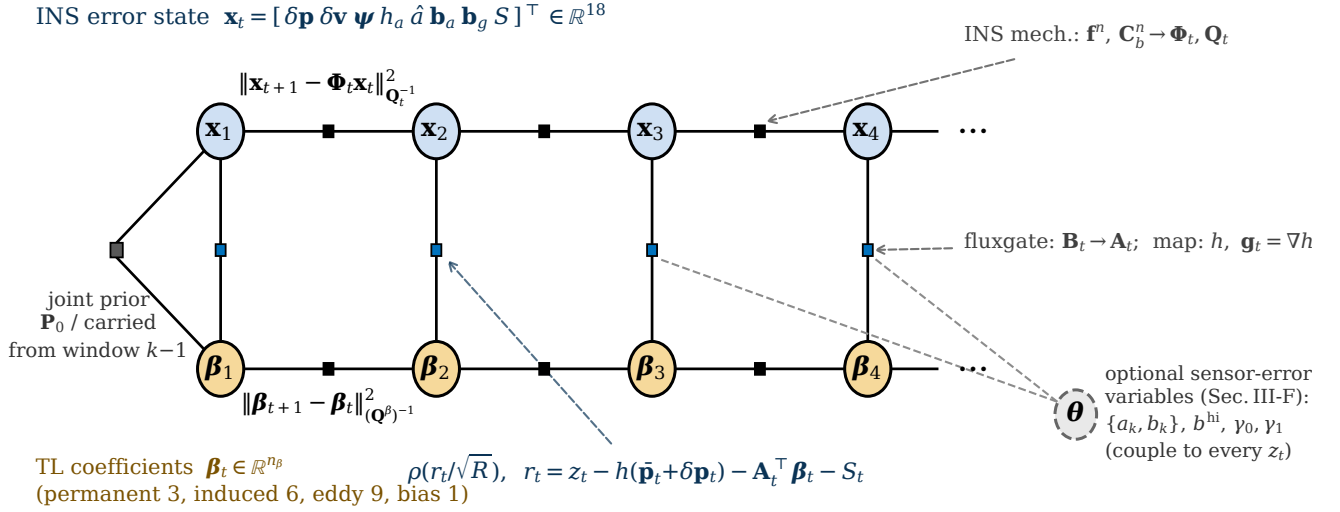


Fig. 2. Factor graph of one window, with every estimated variable and measurement shown. Top: the 18-state INS error chain  $\mathbf{x}_t$  of (1), linked by Pinson–FOGM process factors driven by the INS mechanization. Bottom: the time-varying Tolles–Lawson coefficient chain  $\beta_t$ , linked by random-walk factors  $\mathbf{Q}^\beta$ . Each scalar-magnetometer factor (blue) applies the robust residual of (8), coupling  $\delta \mathbf{p}_t, S_t$ , and  $\beta_t$ ; the joint prior (left) is  $\mathbf{P}_0$  at cold start or the covariance carried from the previous window. Dashed arrows mark known factor parameters (INS mechanization, fluxgate basis  $\mathbf{A}_t$ , map  $h, \nabla h$ ); the dashed node  $\theta$  holds the optional sensor-error variables of Section III-F.

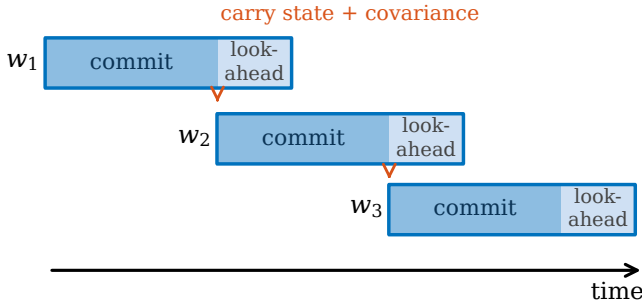


Fig. 3. Fixed-lag sliding window. Each window commits its leading stride and passes the full smoothed state and covariance forward as the prior for the next window; the trailing overlap provides bounded smoother look-ahead.

overlap is a fixed fraction,  $L_o/L_w = 0.3$ , of each window) in exchange for the simpler implementation, and note that the reported DRMS is a comparison of means, unaffected by the covariance bookkeeping. Algorithm 1 summarizes the procedure; its per-window cost is constant in the line length, giving overall linear complexity.

### C. Window design and initialization

Two parameters govern the window: the length  $L_w$  and the overlap  $L_o$ .  $L_w$  sets how much data each compensation estimate sees and thus the effective adaptation time; too short and  $\beta$  is under-determined, too long and it cannot track drift.  $L_o$  sets the look-ahead and hence the commit latency: a committed state has seen  $L_o$  epochs of future data, and the estimator lags real time by  $L_o$ . In the experiments we use  $L_w = 5$  min and  $L_o = 2$  min; the 2 min window in Table III shows that halving  $L_w$  degrades the long-line result, consistent with under-determination. The first window is seeded with a diffuse prior on  $\beta$  (large  $\mathbf{P}_0$  block) so the cold start is not

### Algorithm 1 Fixed-lag sliding-window smoother

- 1: **Input:** measurements  $\{z_t\}$ , TL rows  $\{\mathbf{A}_t\}$ , map  $h$ , window  $L_w$ , overlap  $L_o$ , prior  $(\hat{\chi}_0, \mathbf{P}_0)$
- 2:  $s \leftarrow 1$
- 3: **while**  $s \leq T$  **do**
- 4:   build factor graph on epochs  $[s, \min(s+L_w, T))$  with prior  $(\hat{\chi}_{s-1}, \mathbf{P}_{s-1})$
- 5:   **repeat** ▷ robust Gauss–Newton
- 6:     linearize residuals (8); form weights  $w_t$  from  $\rho$  (11)
- 7:     solve the linearized chain by an iterated RTS pass [41] (equivalently, sparse QR / square-root SAM [32]; Sec. III-E)
- 8:     update  $\chi \leftarrow \chi + \Delta \chi$
- 9:     **until** converged
- 10:   commit epochs  $[s, s+L_w-L_o)$ ; recover  $(\hat{\chi}, \mathbf{P})$  at epoch  $s+L_w-L_o-1$
- 11:    $s \leftarrow s + (L_w - L_o)$
- 12: **end while**
- 13: **Output:** committed trajectory  $\{\hat{\mathbf{p}}_t, \hat{\beta}_t\}$

biased toward zero interference, and the FOGM disturbance state absorbs the initial mismatch while the coefficients converge; subsequent windows inherit the tight carried covariance and converge in a few Gauss–Newton iterations (typically 3–5; Sec. VI-F). This warm-start is why the per-window iteration count stays low despite the nonlinearity of the map term.

### D. Robust kernel and reweighting

Map error, transient interference, and dead-zone measurements produce heavy-tailed residuals that a quadratic cost

over-weights. We use a Huber kernel [43]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \delta, \\ \delta(|u| - \frac{1}{2}\delta), & |u| > \delta, \end{cases} \quad (10)$$

minimized by iteratively reweighted least squares (IRLS): each Gauss–Newton iteration reweights measurement factor  $t$  by

$$w_t = \min(1, \delta/|u_t|), \quad u_t = r_t/\sqrt{R}, \quad (11)$$

so gross outliers are down-weighted toward an absolute-value penalty while inliers keep the efficient quadratic cost. In the implementation the reweighting is applied from the second Gauss–Newton iteration onward (the first iteration is plain least squares) with  $w_t$  clamped to  $[10^{-6}, 1]$ , and  $\delta = 1.345$  gives the standard 95% Gaussian efficiency. Alternatives such as switchable constraints [44] and dynamic covariance scaling [45] fit the same graph interface and could replace (10) unchanged.

### E. Sparse solver

Each Gauss–Newton iteration solves a sparse linear system whose Jacobian has the block-bidiagonal structure of the two chains in Fig. 2 (in the static- $\beta$  limit the coefficient blocks collapse to one shared column). We factor it by column-pivoted QR on the square-root information matrix (square-root SAM [32], [27]), which is numerically stable and exploits the chain sparsity; an iterated RTS pass [41] gives the identical estimate for the Gaussian case and we verify the two agree to within a metre on a real-data segment in Section VI (the check covers the navigation-only chain; the joint TL problem is solved by the iterated-RTS path).

### F. Physical sensor-error factors

The graph representation admits error states that a position-only grid or point-mass estimator cannot carry. Beyond TL we optionally augment  $\chi$  with three physically motivated scalar-magnetometer error variables, each appearing additively in the residual (8), plus one measurement re-weighting:

- 1) *OPM heading error*, modeled as the cosine-harmonic expansion of the optically-pumped (scalar-atomic) magnetometer in the sensor-field angle  $\psi_t = \angle(\hat{\mathbf{e}}, \mathbf{B}_t)$ ,  $b_t^{\text{hdg}} = \sum_{n \in \{1,2,4\}} c_n \cos(n\psi_t)$  [13], [14], with coefficients  $\{c_n\}$  as graph variables; the  $n=1$  term is the vector light shift and  $n=2$  the nonlinear Zeeman effect. Because  $\psi_t$  is known from attitude and IGRF, the term is linear in  $\{c_n\}$  and, as in the Tolles–Lawson case [14], enters the graph as an ordinary linear regressor;
- 2) *Hard-iron bias*  $b_t^{\text{hi}}$ , a constant offset variable;
- 3) *Linear drift*  $b_t^{\text{dr}} = \gamma_0 + \gamma_1 t$ , capturing slow electronics drift;
- 4) *Dead-zone weight*: not a graph variable but a heteroscedastic measurement information  $R_t^{-1} = R^{-1} \max(|\cos \psi_t|, \epsilon)^2$  ( $\epsilon=0.1$ ) that follows the OPM signal-amplitude collapse near the equatorial dead zone (field  $\perp$  optical axis) [13] and de-weights those samples (a de-weighting any estimator could apply; it is included for completeness of the sensor model).

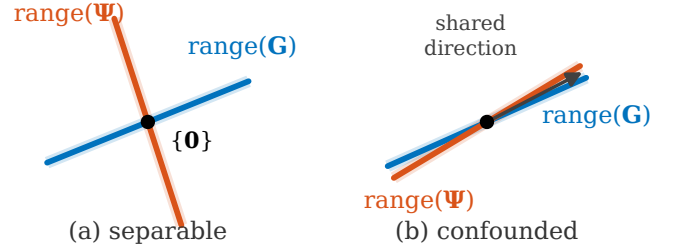


Fig. 4. Observability of joint compensation and navigation (Proposition 1). Left: when the map-gradient subspace  $\text{range}(\mathbf{G})$  and the compensation subspace  $\text{range}(\Psi)$  meet only at the origin, position and compensation are separable. Right: when they share a nonzero direction, a position shift can be masked by a compensation change and observability collapses.

The augmented residual is  $r_t = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \beta_t - S_t - b_t^{\text{hdg}} - b_t^{\text{hi}} - b_t^{\text{dr}}$ . Each term is exogenous to position (Section IV), so it can be identified without stealing map information; their observability is limited only by how much the flight excites the corresponding regressor (e.g. the heading harmonics need attitude variation).

## IV. OBSERVABILITY OF JOINT COMPENSATION AND NAVIGATION

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let  $\mathbf{G}$  have rows  $\mathbf{g}_t^\top$  (the map-gradient/position sensitivity) and let  $\Psi$  have rows  $\mathbf{A}_t^\top$  (the TL regressor). To isolate the position–compensation coupling, restrict (3) to the  $(\delta\mathbf{p}, \delta\beta)$  subspace, giving the window model  $\mathbf{z} = \mathbf{G} \delta\mathbf{p} + \Psi \delta\beta + \boldsymbol{\eta}$  (Fig. 4). This model freezes  $\delta\mathbf{p}$  and  $\delta\beta$  over the window: it is a per-window surrogate for the time-varying system (2), (8) that isolates the position–compensation coupling which a full observability Gramian would mix with the INS dynamics.

*Proposition 1:* A position perturbation  $\delta\mathbf{p}$  is indistinguishable from a compensation adjustment—hence unobservable from map matching over the window—if and only if  $\mathbf{G} \delta\mathbf{p} \in \text{range}(\Psi)$ . Consequently the joint state  $(\delta\mathbf{p}, \delta\beta)$  is observable if and only if the stacked matrix  $[\mathbf{G} \ \Psi]$  has full column rank; equivalently, if and only if all three of the following hold: (i)  $\ker \mathbf{G} = \{0\}$  (the map gradient excites every position direction over the window); (ii)  $\ker \Psi = \{0\}$  (the compensation basis has full column rank); and (iii)  $\text{range}(\mathbf{G}) \cap \text{range}(\Psi) = \{0\}$  (no shared direction). Condition (iii) alone governs the position–compensation *confounding*: when it fails, a position shift with  $\mathbf{G} \delta\mathbf{p} \neq 0$  is reproduced exactly by a compensation change.

*Proof:* Two joint states  $(\delta\mathbf{p}, \delta\beta)$  and  $(\delta\mathbf{p}', \delta\beta')$  produce identical noise-free predictions iff  $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \Psi(\delta\beta' - \delta\beta)$ , i.e. iff  $[\mathbf{G} \ \Psi][(\delta\mathbf{p} - \delta\mathbf{p}')^\top \ (\delta\beta - \delta\beta')^\top]^\top = 0$ . The unobservable subspace is therefore exactly  $\ker[\mathbf{G} \ \Psi]$ , and the joint state is observable iff  $[\mathbf{G} \ \Psi]$  has full column rank; the first statement follows by taking  $\delta\beta'$  free and  $\delta\mathbf{p}' = 0$ . For the three-condition equivalence: if (i) fails, take  $\delta\mathbf{p} \in \ker \mathbf{G} \setminus \{0\}$ , giving the kernel element  $(\delta\mathbf{p}, 0)$ ; if (ii) fails, take  $\delta\beta \in \ker \Psi \setminus \{0\}$ , giving  $(0, \delta\beta)$ ; if (iii) fails, take  $\mathbf{G} \delta\mathbf{p} = \Psi \delta\beta \neq 0$ , giving  $(\delta\mathbf{p}, -\delta\beta)$ . Conversely, let  $(\delta\mathbf{p}, \delta\beta) \neq (0, 0)$  lie



in the kernel, so  $\mathbf{G} \delta \mathbf{p} = -\Psi \delta \beta$ . If  $\mathbf{G} \delta \mathbf{p} = \mathbf{0}$  then either  $\delta \mathbf{p} \neq \mathbf{0}$  violates (i) or  $\delta \beta \neq \mathbf{0}$  violates (ii); if  $\mathbf{G} \delta \mathbf{p} \neq \mathbf{0}$  it is a nonzero shared direction, violating (iii). ■

*Corollary 1 (Exogeneity and separability):* Suppose  $\mathbf{A}_t$  is a function of aircraft attitude/fluxgate alone—exogenous to position. Then no column of  $\Psi$  is generated by the map gradient along the trajectory, and condition (iii) holds for generic trajectories and maps (in particular, it requires the window to satisfy  $N \geq \dim \text{range}(\mathbf{G}) + \dim \text{range}(\Psi)$  and no  $\Psi$  column to be a linear combination of the map-gradient time series). Position-compensation *separability* is then independent of the basis dimension: enlarging an exogenous basis can slow the coefficient estimate through (ii) but cannot mask position. If instead a compensation feature contains the map value  $h(\mathbf{p})$  (for example the scalar magnetometer itself), its column lies in  $\text{range}(\mathbf{G})$  by construction, forcing the intersection nonzero and collapsing observability.

*Remark 2 (TL rank deficiency as gauge freedom):* Condition (ii) fails for the classical TL basis itself: with unit direction cosines, the three induced diagonal columns sum to  $B_t(u^2 + v^2 + w^2) = B_t$ , collinear with the constant-bias column—the long-known collinearity of TL regression [23], [25]. This is a *gauge freedom* of the compensation, not a navigation defect: any  $\delta \beta \in \ker \Psi$  leaves  $c_t = \mathbf{A}_t^\top \beta_t$ —and hence every prediction and the position estimate—unchanged. The prior  $\mathbf{P}_0$  and the random-walk factors select one representative of each gauge orbit. Separability of position from compensation is governed by (iii) alone and is unaffected.

Proposition 1 is exact but non-constructive: it does not yield a reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection (in-sample basis  $R^2$ , out-of-sample prediction, and gradient-alignment indices) did not generalize across estimators or lines. The actionable consequence is Corollary 1: drive the compensation from exogenous features, which every estimator in Section VI does.

*Remark 3:* The condition is geometric, not statistical: it depends on the ranges of  $\mathbf{G}$  and  $\Psi$  over the window, hence on the flight path and the local map gradient, not on noise levels. It explains two failure modes seen in practice. First, over a locally flat map  $\mathbf{G} \approx \mathbf{0}$  and condition (i) fails: position is unobservable regardless of the compensation—a map-information floor that bounds every estimator, and that we observe on the shortest line in Section VI. Second, an *endogenous* feature such as the scalar magnetometer reading itself makes a  $\Psi$  column collinear with  $\mathbf{G}$ , so the compensator silently absorbs the map signal; excluding such features (Section V) is necessary for the baseline to converge.

## V. REIMPLEMENTED ONLINE EKF+TL+NN BASELINE

To compare against [12] on equal footing we *reimplement*, rather than merely cite, the online EKF+TL+NN cold start. The filter augments the Pinson error state with the online TL coefficients  $\beta$  and the weights of a small feed-forward network, all propagated as EKF states and updated online against (3). Because neither the network architecture nor the natural-gradient training design of [12] is released, ours is a reimplementation of the published design principles, not a line-for-line replication. Naively initialized, the reimplementation

diverged; three fixes, each addressing a distinct and independently diagnosed failure, were required.

(i) *Bias-free network with DC initialization.* The map returns the anomaly only, whereas the raw scalar reading includes a large constant offset. Left in the regression target, this offset drove the network output bias to  $\sim 5 \times 10^4$  nT and the filter to NaN. We remove the mean by a short initial window, initializing the network output bias to the median measured-minus-predicted offset and scaling the target by its standard deviation.

(ii) *Exogenous network inputs only.* Using the scalar magnetometer as a network input creates exactly the endogenous feature of the Remark above: the network learns to reproduce the map and horizontal position collapses (we observed  $> 5$  km error with a small measurement residual). Restricting the inputs to the attitude-derived TL basis restores observability, in direct agreement with Corollary 1.

(iii) *De-trusting the map through the transient.* During the cold-start transient the compensation is grossly wrong, so the map residual is unreliable; we inflate the measurement variance and the coefficient process noise for the warm-up, then relax them.

Concretely, the reimplemented filter augments the navigation state with the permanent TL group and a single-hidden-layer network of eight units acting on that group, giving 41 network-weight states carried in the EKF; the FOGM disturbance state uses a  $(\sigma, \tau)$  of (3 nT, 180 s) and the measurement variance is  $12^2 \text{ nT}^2$  after warm-up. The DC-removal step measures an interference offset of, e.g.,  $-1724$  nT on Mag 4 and  $+310$  nT on Mag 5, which if left in the target is what drove the untreated filter to divergence. With these three fixes the filter reaches 40.0/17.5 m on Mag 4/5 (Table III): on Mag 4 this lies inside the 37 m to 58 m band spanned by the reported TL-only and TL+NN results, while on Mag 5 it is 2.5 m to 3.5 m above the reported values (14 m to 15 m)—a functioning, close, but not exact reimplementation. Its late-half residual falls to 117 nT to 231 nT, and we use it as the baseline. We stress that the network only *compensates*; the fixes are what make it observable jointly with position, which is the same requirement Corollary 1 places on our own compensation features.

## VI. EXPERIMENTS AND RESULTS

### A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [8]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically-pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines accordingly. Flt1007 is the *free-flight* mission—the aircraft flies freely within the mapped Renfrew ( $\approx 400$  m) and Eastern/Perth ( $\approx 800$  m) areas—so its lines are the closest to a navigation task and are what prior work [8], [4], [12] reports; 1007.06 is our primary line for that reason. Flt1003 is a crustal-field *measurement* flight (survey traverses at 400 and 800 m); its lines 1003.02 and

TABLE I

ESTIMATOR SETTINGS USED FOR ALL COLD-START RESULTS (VALUES FROM THE RELEASED CODE). THE SAME  $\mathbf{P}_0$ ,  $\mathbf{Q}$ ,  $\mathbf{R}$ , AND TL SETTINGS DRIVE BOTH THE CAUSAL EKF AND THE WINDOW SMOOTHER.

| Group                  | Parameter                              | Value                                           |
|------------------------|----------------------------------------|-------------------------------------------------|
| Measurement            | noise std. $\sqrt{R}$                  | 12 nT                                           |
| FOGM $S$               | $(\sigma_S, \tau)$                     | 3 nT, 180 s                                     |
| Initial $\mathbf{P}_0$ | pos. / alt. std.                       | 0.1 / 1.0 m                                     |
|                        | velocity std. (per axis)               | 1.0 m s <sup>-1</sup>                           |
| TL prior / walk        | $\mathbf{P}_0^\beta, \mathbf{Q}^\beta$ | $\mathbf{I}(1.0)^2; \mathbf{I}(1.0)^2 \Delta t$ |
| TL basis               | terms (App. B)                         | P+I+E+bias (19)                                 |
| Window                 | $L_w / L_o / \Delta t$ [s]             | 300 / 90 / 0.1                                  |
| Robust                 | Huber $\delta$                         | 1.345                                           |
|                        | IRLS                                   | 2nd iter. on, $w \in [10^{-6}, 1]$              |
| Gauss–Newton           | max. iter. / tol.                      | 5 / $10^{-4}$                                   |

1003.08 are straighter than the free-flight legs. Flt1006 is a *calibration-maneuver* flight, so we set its single short line aside (Section VI-D), and the map-building flights (Flt1004–1005) are not used. Two honest points remain. First, none of these is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry more attitude variation, which stresses compensation but also *aids* TL observability by exciting the regressor (Section IV), so performance on a pure operational cruise is not directly established here and is future work. Second, the lines span two altitude regimes—four at the  $\approx 400$  m survey-drape altitude and 1007.02 at  $\approx 800$  m, where upward continuation attenuates the anomaly. Map matching uses the Eastern (“E”) and Renfrew (“R”) high-resolution anomaly grids, upward-continued to flight altitude with the IGRF core field restored. The metric is horizontal distance-RMS (DRMS) of position error after a 10 min warm-up. *Cold start* means the TL coefficients are initialized to zero with no calibration flight. For every comparison the EKF and FGO estimators use identical dynamics, map interpolation, and noise settings (Table I); the window smoother additionally applies the Huber kernel (10), whose isolated contribution is reported separately in Table III (37.0→32.6 m on Mag 4), so the reader can attribute the remaining margin to the estimator structure. Unless noted, the window smoother uses  $L_w = 5$  min and  $L_o = 90$  s;  $L_w$  was selected on line 1007.06 (Fig. 8) and then held fixed for every other line, so the breadth results of Table IV are not per-line tuned. All numbers are produced by continuous-integration scripts on the code branch and are reproducible from it.

### B. Batch estimator on real data

We first validate the estimator on line 1003.02 (Flt1003, 63 min) with the compensated stinger magnetometer, where the interference is already removed and the comparison isolates the estimator. The batch factor graph improves horizontal DRMS from the EKF’s 28.3 m to 14.0 m (Huber kernel), against 114.6 m free-inertial—a 51 % reduction over the EKF. Fig. 5 shows the anomaly map, the flight line, and a mid-flight zoom in which the free-inertial track visibly departs from truth while the EKF and FGO estimates follow it. Fig. 6 plots the horizontal error over the line: the FGO estimate (blue) stays below the EKF (red) for essentially the entire flight, the shaded

TABLE II

BATCH ESTIMATOR ON LINE 1003.02 (COMPENSATED STINGER); HORIZONTAL DRMS [M]. FREE-INERTIAL REFERENCE 114.6 m. THE RTS AND GN/QR SOLVERS AGREE, CONFIRMING A COMMON MAP ESTIMATE.

| Segment   | Estimator               | DRMS        |
|-----------|-------------------------|-------------|
| full line | EKF                     | 28.3        |
| full line | FGO (RTS)               | 14.3        |
| full line | <b>FGO (RTS, Huber)</b> | <b>14.0</b> |
| 10 min    | EKF                     | 47.9        |
| 10 min    | FGO (RTS)               | 18.4        |
| 10 min    | FGO (GN/QR)             | 19.4        |
| 10 min    | FGO (GN/QR, Huber)      | 18.4        |

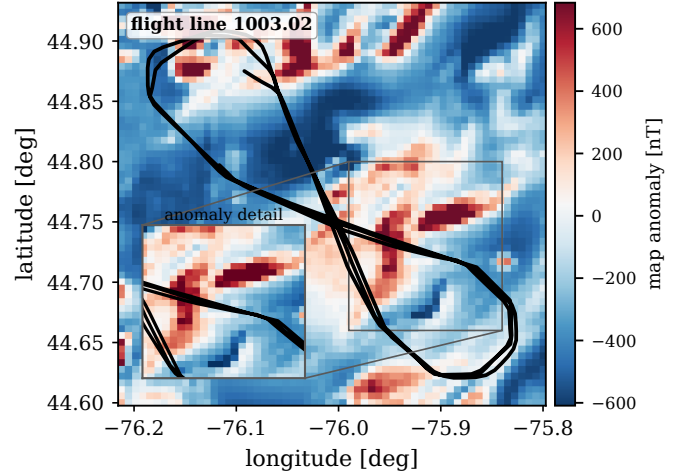


Fig. 5. Eastern anomaly map and flight line 1003.02, providing geographic context; the inset zooms the fine anomaly structure the map-matching factor exploits along the line. The estimator comparison is deferred to Figs. 6–7: at the survey ground speed a metre-level track error is smaller than one 12 s sample and cannot be resolved on a map.

region marking the improvement. Fig. 7 gives the empirical error distribution: the FGO curve reaches its 95th percentile at less than half the error of the EKF, and both bound the free-inertial error by an order of magnitude. On a 10 min segment the iterated-RTS and sparse-GN/QR solvers agree to within 1 m (Table II), confirming that both numerically reach the same MAP estimate as predicted by Lemma 1. The Huber kernel changes the batch result by less than 0.3 m here, as expected for the already-compensated stinger sensor whose residuals are close to Gaussian; its value appears in the cold-start cases below, where transient outliers are large.

### C. Cold-start line vs. online EKF+TL+NN

The central comparison is the cold start on uncompensated cabin magnetometers, where the interference is present and must be learned online. Table III and Fig. 9 report the primary evaluation line 1007.06 (87 min, full length) on Mag 4 and Mag 5. A static-TL batch forces a single coefficient vector over the whole line; it underfits the time-varying interference and degrades to 123.7/68.1 m. Allowing the compensation to adapt with a 5 min window recovers it to 37.0/15.1 m, and the Huber kernel further trims transient outliers to 32.6/14.2 m. Our reimplemented online EKF+TL+NN reaches 40.0/17.5 m



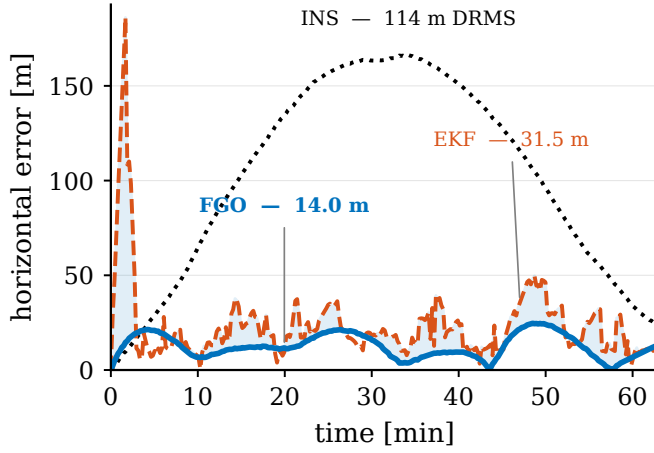


Fig. 6. Horizontal position error on line 1003.02. The batch factor graph (FGO, blue) holds a tighter bound than the EKF (red) across the line; the shaded band marks the FGO improvement, and free inertial (INS) diverges. The labelled DRMS values are computed over the full line, without the warm-up exclusion of Table II.

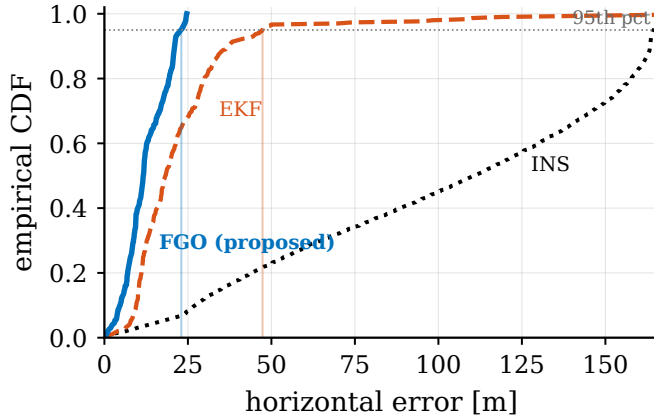


Fig. 7. Empirical CDF of horizontal error on line 1003.02 (full line, as in Fig. 6). The proposed FGO reaches its 95th percentile at less than half the EKF's error; both bound the free-inertial error by an order of magnitude.

(Section V details how close this is to the reported operating point), while the reported TL-only EKF is weaker still on Mag 4. The window smoother beats the reimplemented NN-augmented filter on both sensors *without any neural network*, using only the linear TL basis carried as graph variables; against the values reported in [12] it is 4.4 m better on Mag 4 and 0.2 m behind on Mag 5. This comparison is on the single line 1007.06—the line [12] evaluates—so the NN baseline breadth across the other four lines remains untested. We attribute the margin to the window look-ahead: the coefficients are identified from data on both sides of each committed epoch, whereas the EKF must commit with past data only.

The window length itself trades under-determination against staleness. Fig. 8 plots DRMS at the three lengths evaluated on this line: the 2 min window under-determines the coefficients, the full-line static batch cannot track their drift, and the 5 min window is the best of the three. The tradeoff mirrors the expressiveness discussion of Section IV and makes  $L_w$  a physically interpretable knob rather than a free hyperparameter,

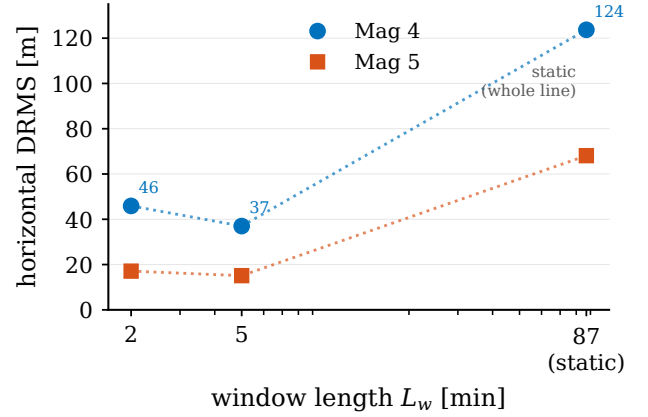


Fig. 8. Window length versus accuracy on line 1007.06 (cold start), at the three evaluated lengths. The 5 min window balances the under-determined 2 min window against the drift-blind static batch (full line).

TABLE III  
COLD START, LINE 1007.06, UNCOMPENSATED CABIN MAGNETOMETERS;  
DRMS [M]. INS 318 m; EKF ON COMPENSATED MAG 1  $\approx$  20 m.

| Method (no NN unless noted)             | Mag 4       | Mag 5       |
|-----------------------------------------|-------------|-------------|
| FGO-online, static batch TL             | 123.7       | 68.1        |
| FGO-online, window 2 min                | 45.9        | 17.1        |
| FGO-online, window 5 min                | 37.0        | 15.1        |
| <b>FGO-online, window 5 min + Huber</b> | <b>32.6</b> | <b>14.2</b> |
| EKF+TL+NN, reimplemented [12]           | 40.0        | 17.5        |
| EKF+TL+NN, reported [12]                | 37          | 14          |
| EKF+TL-only, reported [12]              | 58          | 15          |

though a denser sweep (and other lines, where the balance can shift toward longer windows) is left for the revision.

#### D. Breadth across lines, flights, and maps

Table IV runs the same cold-start TL model on five long survey lines spanning three flights and two maps, against *two* causal baselines: the plain online-TL EKF, and the reimplemented EKF+TL+NN of Section V, the strong causal method that a fair comparison demands.

The two baselines behave very differently, and the contrast is the point. The plain online-TL EKF is fragile: on the noisy Mag 4 it diverges from the cold start to tens of kilometres on three lines ( $17\text{--}42 \times 10^3$  m) and runs off the map on a fourth. The neural network fixes exactly this—the EKF+TL+NN stays finite on all 10 cases, confirming that its residual network is what buys cold-start robustness for the causal filter. *The window smoother matches that robustness without any neural network*—it is bounded on all 10—and improves on accuracy: on the four full-length lines (63 min to 87 min, eight mag-cases) it is the best of the three on *all eight*, beating the NN-augmented filter by 4 m to 76 m (Fig. 10). The fifth line, 1006.08, is from a compensation-maneuver flight and is only 14 min long, so the 10 min warm-up leaves a 4 min evaluation window over which none of the three DRMS values is reliable; we retain it in Table IV because it still exhibits the causal-EKF divergence ( $1.7 \times 10^4$  m on Mag 4) and the map-information floor that makes every method weak, but we exclude its two

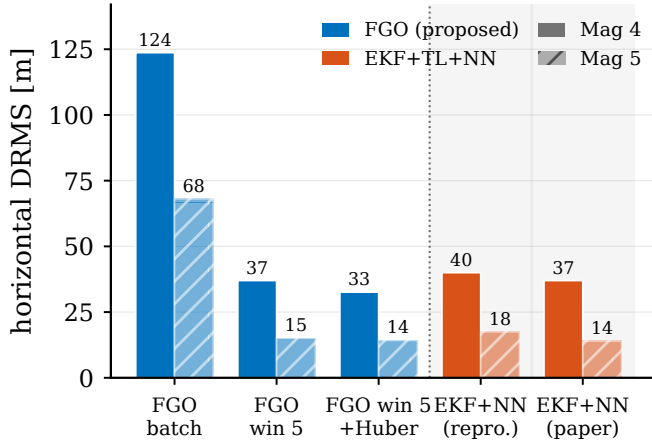


Fig. 9. Line 1007.06 cold start on the uncompensated cabin magnetometers. The NN-free fixed-lag window (left of the divider) compares against the reimplemented and reported EKF+TL+NN (shaded, right).

cases from the count and draw no accuracy conclusion from its ordering (which is why the NN’s 34.6 m on Mag 5, against 117–122 m for the TL-only methods, is not evidence of an NN accuracy advantage).

Two effects in the reliable cases are worth naming. First, the EKF+TL+NN follows the reproduced design in feeding the network only the *permanent* TL group, whereas the online-TL EKF and the window smoother both carry the full permanent-induced-eddy-bias basis; on the well-conditioned Mag 5 those linear higher-order terms matter, so the NN filter can fall *behind even the plain EKF* (1003.02: 33.8 vs 28.1 m) when its network has not yet learned what the dropped linear terms would have supplied directly. Second, and relatedly, the network’s extra online-learned states add estimation variance where the linear basis already suffices, a cost the NN-free window avoids. Both effects favor a linear compensation carried in a smoother over a learned one carried in a filter, on this benchmark. One caveat on provenance: the EKF+TL+NN is our reimplementation, matching the reported 1007.06 operating point to within a few metres (Section V), but the original’s per-line numbers on the other lines are unpublished, so the comparison is against a faithful—not the authors’—filter.

#### E. Sensor-error factors (simulation)

To exercise the sensor-error factors of Section III we inject the optically-pumped-magnetometer error model of [13], [14] into a simulated flight (free-inertial reference 31.1 m): a cosine-harmonic heading error at orders  $\{1, 2, 4\}$  (7.6 nT RMS), a fluxgate hard-iron bias, a linear drift, and the equatorial-dead-zone noise amplification  $1/\max(|\cos \psi|, \epsilon)$ . We then enable the corresponding graph variables cumulatively; Table V reports the result. With no sensor-error states the estimate is *worse* than free inertial (42.9 m): the unmodeled error corrupts the map aiding, exactly the mechanism of Proposition 1. Enabling all four factors reduces the error to 6.2 m, an 86% reduction, monotone in this order. Because the factors are enabled cumulatively in a single order, Table V

TABLE IV  
BREADTH: COLD-START CABIN MAGNETOMETERS, DRMS [M], ON THE SGL 2020 NAVIGATION-USABLE LINES. “PURPOSE”: FF = FREE-FLIGHT (FLT1007), SV = SURVEY TRAVERSE (FLT1003), CAL = CALIBRATION-MANEUVER FLIGHT (FLT1006). “DIV.” = DIVERGENCE > 10 km; “ERR.” = RUN LEFT THE MAP. BOLD = BEST OF THE THREE BASELINES. THE CALIBRATION LINE 1006.08 (†) IS ONLY 14 min (A 4 min WINDOW AFTER WARM-UP), UNRELIABLE AND EXCLUDED FROM THE COUNT.

| Line                                                | Purp. | Map | alt [km] | Mag | EKF online | EKF+TL+NN | FGO win      |
|-----------------------------------------------------|-------|-----|----------|-----|------------|-----------|--------------|
| 1007.06                                             | FF    | R   | 0.4      | 4   | 46.7       | 42.2      | <b>32.7</b>  |
| 1007.06                                             | FF    | R   | 0.4      | 5   | 17.8       | 17.8      | <b>13.8</b>  |
| 1007.02                                             | FF    | E   | 0.8      | 4   | div.       | 114.8     | <b>38.6</b>  |
| 1007.02                                             | FF    | E   | 0.8      | 5   | 31.6       | 29.3      | <b>14.5</b>  |
| 1003.02                                             | SV    | E   | 0.4      | 4   | div.       | 99.1      | <b>42.6</b>  |
| 1003.02                                             | SV    | E   | 0.4      | 5   | 28.1       | 33.8      | <b>21.7</b>  |
| 1003.08                                             | SV    | R   | 0.4      | 4   | err.       | 46.4      | <b>26.1</b>  |
| 1003.08                                             | SV    | R   | 0.4      | 5   | 21.1       | 18.8      | <b>12.4</b>  |
| <i>calibration flight (†, excluded from count):</i> |       |     |          |     |            |           |              |
| 1006.08†                                            | CAL   | E   | 0.4      | 4   | div.       | 1371      | 193.9        |
| 1006.08†                                            | CAL   | E   | 0.4      | 5   | 117.5      | 34.6      | 122.0        |
| FGO best (navigation lines)                         |       |     |          |     |            |           | <b>8 / 8</b> |

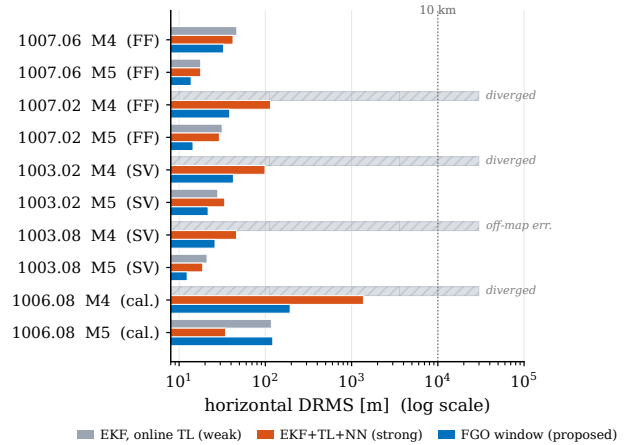


Fig. 10. Breadth study (Table IV) on a log DRMS axis: weak online-TL EKF (gray, hatched where it diverges beyond 10 km), strong EKF+TL+NN (red), and the NN-free FGO window (blue). The NN keeps the causal filter bounded; the window matches that and is best of the three on all 8 full-length cases (the 14 min line 1006.08 is excluded as its post-warm-up window is too short to compare).

attributes marginal gains only for that order: the largest single-step drop comes from the heading harmonics (42.9 → 18.9 m), followed by the dead-zone weighting (→ 15.5 m, now load-bearing because the injected dead-zone noise is heteroscedastic), the hard-iron bias (→ 8.1 m), and the linear drift, whose rate the estimator recovers cleanly ( $\hat{\gamma}_1 = 0.018$  versus a true 0.020). The heading coefficients themselves are poorly identified ( $\hat{c}_1 = 3.8$  versus a true 10): level flight sweeps the sensor-field angle by only about  $12^\circ$ , so the harmonic regressor is barely excited—an observability property of the trajectory, consistent with the Remark of Section IV, that a maneuvering flight would relieve, and the gain to navigation comes from the fitted combination over the observed angle range, not from the individual coefficients. The Huber kernel gives similar values (within 2 m; it helps most at the dead-zone

TABLE V

SENSOR-ERROR FACTORS (SIMULATION), ENABLED CUMULATIVELY; HORIZONTAL DRMS [M]. FREE-INERTIAL REFERENCE 31.1 m. VALUES FOR THE NON-ROBUST SOLVER; THE HUBER SOLVER AGREES WITHIN 2 m.

| Graph variables enabled       | DRMS       |
|-------------------------------|------------|
| none (map aiding only)        | 42.9       |
| + heading harmonics {1, 2, 4} | 18.9       |
| + dead-zone weight            | 15.5       |
| + hard-iron bias              | 8.1        |
| + <b>linear drift (full)</b>  | <b>6.2</b> |

step). Two caveats bound this study: it is simulation-only, and the injected errors share the estimator’s functional form, so it demonstrates that the graph formulation can carry and recover such states, not that these models capture real sensor behavior.

#### F. Computational considerations

Each window is a sparse nonlinear least-squares problem of fixed size  $O(L_w(n + m))$ ; the block-bidiagonal Jacobian of the two chains (Fig. 2) is factored in  $O(L_w)$  by the square-root solver [32], and the number of Gauss–Newton iterations is small (typically 3–5) because the problem is mildly nonlinear away from observability collapse. The per-window cost is therefore constant in the line length and the overall complexity is linear in the number of epochs, matching a filter’s asymptotic cost while retaining the look-ahead. Memory is bounded by the window, not the flight. The one-time overhead relative to an EKF is the window factorization and the IRLS reweighting, which in our unoptimized research implementation runs comfortably faster than real time on the SGL sampling rate.

## VII. DISCUSSION

#### A. Why the window helps

The advantage is not smoothing per se, which equals the RTS smoother [41] in the linear–Gaussian case, but the joint windowed estimation of a *time-varying* compensation together with navigation. A causal filter must commit each state online with no future context; under a large uncompensated cold-start residual it has only the position states with which to explain the mismatch, attributes it to position, and—as Table IV shows on the noisy Mag 4—can diverge to tens of kilometres. The fixed-lag window defers commitment by  $L_o$  epochs, so the compensation is identified from data on both sides of each committed state before that state is fixed; later-observable calibration then protects earlier navigation. This is the same structural reason that batch and incremental smoothers outperform filtering in visual–inertial estimation [38], [40], here applied to the compensation problem.

The breadth study separates two things a single-line comparison conflates: robustness and accuracy. Robustness is not unique to the window—the neural network also keeps the causal EKF bounded on every cold start (Table IV), so the naive claim that “causal filters diverge” is false for the state-of-the-art filter. What is distinctive is that the window smoother reaches that same bounded regime *without* the network, its

TABLE VI

ESTIMATOR FAMILIES FOR COLD-START MAP-MATCHING COMPENSATION. ✓ = YES, ✗ = NO, (∼) = LIMITED.

| Estimator                       | Future data | Comp. state | Non-Gauss. | Bounded latency |
|---------------------------------|-------------|-------------|------------|-----------------|
| Causal EKF [4]                  | ✗           | (∼)         | ✗          | ✓               |
| Particle / grid filter [7]      | ✗           | ✗           | ✓          | ✓               |
| Online EKF+TL+NN [12]           | ✗           | ✓           | (∼)        | ✓               |
| Batch FGO                       | ✓           | ✓           | ✓          | ✗               |
| <b>Proposed (fixed-lag FGO)</b> | ✓           | ✓           | ✓          | ✓               |

online-training design, or the associated verification burden, and then improves on accuracy: it is best of the three on all 8 full-length cases. The gain over the strong baseline is real but should not be over-read—our EKF+TL+NN is a reimplemention, and part of its deficit is that its online-learned states add variance on the well-conditioned sensor where the linear TL basis already suffices. The defensible summary is that a linear, NN-free estimator, given future context through the window, is sufficient to both survive the cold start and match or beat an NN-augmented causal filter on this benchmark. Table VI positions the estimator families along the axes that matter for cold-start compensation.

#### B. Handling the non-Tolles–Lawson remainder

A fair question is how the estimator copes with the large fraction of the SGL 2020 cabin-sensor interference that the linear TL basis cannot span (platform currents and other non-attitude sources), given that it carries no neural network for the purpose. It does not model that remainder as the network does; it absorbs it through three linear–Gaussian and robust mechanisms. First, the *time-varying* coefficients  $\beta_t$ : a static TL fit leaves the slowly drifting part of the current signature in the residual, whereas the per-window relinearization lets  $\beta_t$  follow it, recovering the low-frequency, maneuver-correlated component a calibration flight cannot. Second, the FOGM disturbance state  $S_t$ , a first-order Gauss–Markov process that absorbs the temporally correlated but attitude-unexplained residual—the linear–Gaussian analogue of a learned residual, though only a scalar AR(1) and so far less expressive than a network. Third, the Huber kernel, which treats the heavy-tailed transient currents that neither term explains as outliers to be down-weighted rather than fit, so they cannot corrupt position. The honest summary is a trade: the network buys a lower compensation residual, while the window+FOGM+robust combination buys navigation robustness *despite* a higher residual—which is why our uncompensated-Mag 4 DRMS (tens to hundreds of metres) is far above survey-grade compensation yet still bounds the INS drift. The factor graph also leaves an obvious upgrade path that keeps the formulation intact: a learned residual can be added as an additional factor (for example a Gaussian-process or small-network term on the same measurement), giving the network’s expressiveness within the graph rather than inside a filter—which we leave to future work.

### C. The role of observability

Proposition 1 delimits when the joint problem is well-posed at all: the map-gradient and compensation subspaces must intersect only trivially. Its two corollaries organize the empirical behavior. Exogeneity (Corollary 1) is why excluding the scalar magnetometer from the network inputs was necessary for the baseline to converge, and why the attitude-only TL basis is safe at any dimension. The map-information floor of the Remark is why the short compensation-flight line 1006.08 is hard for every estimator—there is simply little independent  $\mathbf{g}_t$  to exploit over its brief window, which is why we set it aside rather than draw a conclusion from it. We stress that the condition is geometric and predictive only through these qualitative rules; our attempts at a scalar screening statistic did not generalize, which we report as a negative result rather than overclaim.

### D. Deployment considerations

The estimator is compatible with real-time operation. Its committed output lags real time by the overlap  $L_o$  (here 2 min), a latency that is acceptable for the map-aiding role—the INS provides the high-rate, zero-latency solution, and the smoother periodically corrects it—but that a designer trades against accuracy by choosing  $L_o$ . Because each window is bounded, both computation and memory are constant per commit and independent of flight duration, unlike a growing batch. The factor interface is modular: the compensation, sensor-error, and map factors are added or removed without touching the solver, so the same implementation covers the calibrated (frozen  $\beta$ ), cold-start (adaptive  $\beta$ ), and augmented-sensor cases. Integration with an operational system therefore amounts to feeding the INS mechanization, fluxgate, and scalar magnetometer streams into the graph and reading back the corrected position at the commit rate. The main tuning burden is the two window parameters and the process noise  $\mathbf{Q}^\beta$ , all of which have clear physical interpretations.

### E. Limitations

Several limitations bound the present study, and we state them plainly. The real-data results are single-run DRMS on five lines; Monte-Carlo statistics over initializations and the full SGL line set remain to be run. We compare against a causal online-TL EKF and a reimplemented online EKF+TL+NN, but not against a like-for-like grid or particle-filter map-matching baseline [16], [7], which would sharpen the estimator-structure claim. The sensor-error factors are validated in simulation only; a flight demonstration with genuine optically-pumped-magnetometer heading error is future work. Most importantly, although the primary line comes from the free-flight mission (Flt1007) and is the community's navigation benchmark, none of the SGL 2020 lines is a dedicated long, straight-and-level operational transit; their attitude content both stresses compensation and, through the excitation of the TL regressor, aids its observability, so performance on a pure operational cruise with little attitude variation is not directly established here and is the most important next

step. Finally, short lines remain hard for every method, an information limit rather than an estimator one. None of these affects the central, reproducible finding: a linear, NN-free window smoother is the best of three cold-start estimators on all 8 full-length cold-start cases—bounded where the plain causal filter diverges, and better than an NN-augmented causal filter without using a network.

## VIII. CONCLUSION

Casting online Tolles–Lawson aeromagnetic compensation as time-varying factor-graph variables in a fixed-lag sliding-window smoother yields a neural-network-free cold-start calibrator that, on real SGL 2020 data, matches or beats a reimplemented online EKF+TL+NN on the primary line and generalizes across five lines, three flights, and two maps, remaining bounded where the causal EKF diverges. An exact observability condition and its exogeneity corollary explain the behavior and guide the choice of compensation features.

## APPENDIX A

### PINSON ERROR-STATE DYNAMICS

The navigation-error block of  $\Phi_t$  in (2) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors  $(\delta\mathbf{p}, \delta\mathbf{v}, \psi)$  and sensor biases  $(\mathbf{b}_a, \mathbf{b}_g)$ , the continuous error dynamics are

$$\begin{aligned}\dot{\delta\mathbf{p}} &= \mathbf{F}_{pp} \delta\mathbf{p} + \delta\mathbf{v}, \\ \dot{\delta\mathbf{v}} &= \mathbf{F}_{vp} \delta\mathbf{p} + \mathbf{F}_{vv} \delta\mathbf{v} - [\mathbf{f}^n \times] \psi + \mathbf{C}_b^n \mathbf{b}_a, \\ \dot{\psi} &= \mathbf{F}_{\psi p} \delta\mathbf{p} + \mathbf{F}_{\psi v} \delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n \times] \psi - \mathbf{C}_b^n \mathbf{b}_g,\end{aligned}\tag{12}$$

where  $\mathbf{f}^n$  is the specific force,  $\mathbf{C}_b^n$  the body-to-nav rotation,  $\boldsymbol{\omega}_{in}^n$  the inertial rate,  $[\cdot \times]$  the skew operator, and the  $\mathbf{F}_{\bullet\bullet}$  blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state  $S$  with correlation time  $\tau$ . The discrete  $\Phi_t = \exp(\mathbf{F}_t \Delta t)$  and  $\mathbf{Q}_t$  are obtained by standard van Loan discretization [42]. Only  $\mathbf{g}_t^\top$ , the position row of the measurement Jacobian (9), couples this block to the map.

## APPENDIX B

### TOLLES–LAWSON BASIS

The basis row  $\mathbf{A}_t$  of (6) is built from the fluxgate direction cosines  $\hat{\mathbf{b}} = [u \ v \ w]^\top$  (with  $u^2 + v^2 + w^2 = 1$ ) and the scalar field magnitude  $B$  [9], [23]. The permanent moment contributes the three cosines  $(u, v, w)$ ; the induced magnetization, symmetric in the field, contributes the six products  $B(u^2, uv, uw, v^2, vw, w^2)$ ; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products  $B(u\dot{u}, u\dot{v}, \dots, w\dot{w})$ . Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and  $\mathbf{c}_t = \mathbf{A}_t^\top \beta_t$  is linear in the coefficients  $\beta_t$ —the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes  $\beta$  is unavailable, which is precisely why we carry it as a graph variable.

## ACKNOWLEDGMENT

The experiments use the publicly released SGL 2020 challenge dataset [8].

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